



Encode



Shuffle



Analyze



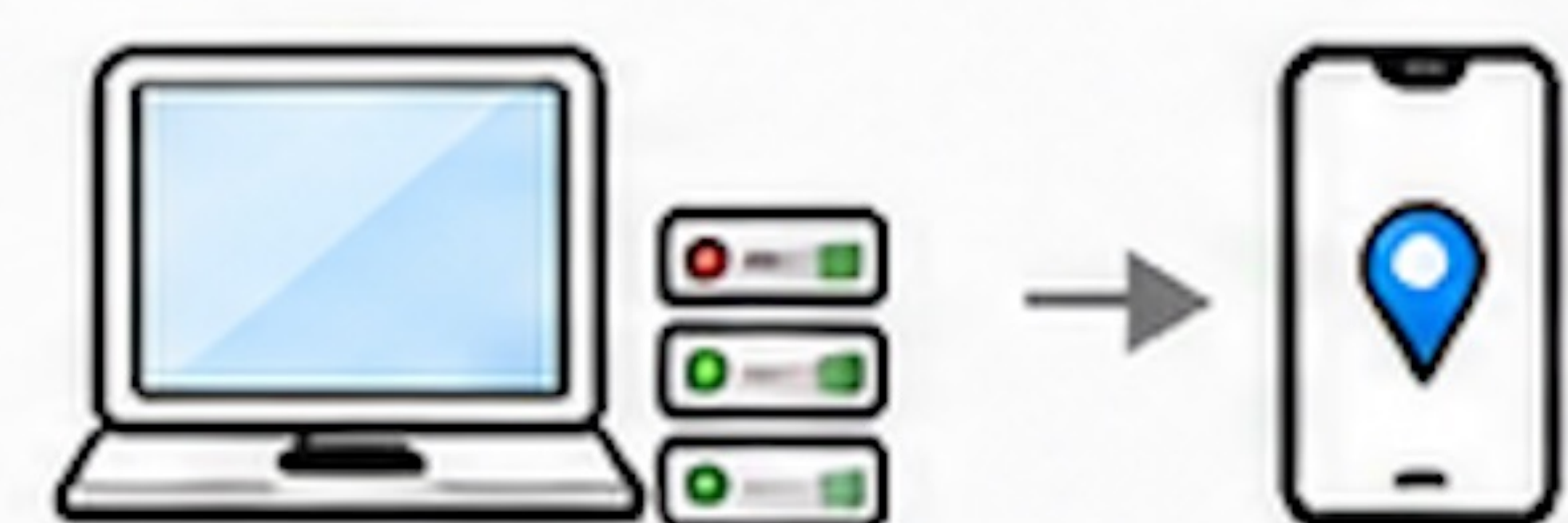
Trusted Setup

- Groth16 setup for the main proof
- $pk \rightarrow$ Clients
- $vk \rightarrow$ Shuffler



Enrollment Attestation Authority (EA)

- ✓ federated enrollment anchors
- ✓ threshold attestation
- ✓ attested enrollment metadata
- ✓ mobility-capable parameters
- ✓ mode commitment
- ✓ secret commitment



Enrollment / Warmup of the aggregation path

- warmup updates state
- not aggregated
- $N_{\text{turn}} = \text{max}(K, 2)$



data flow



trust / assumption



check passed

Stage 1: Client-Side Processing



1 Local trajectory state

- hidden coordinates
- timestamp
- previous / anchor commitments



2 Payload-binder construction

- derive hidden primary cell
- construct public payload binders
- bind the payload to the committed primary



3 Commitment-chain update

- location commitments
- commitment chain
- binds state / round / payload context



4 Proof generation

- ✓ per-step trajectory continuity
- ✓ ADNC window bound
- ✓ hidden primary-cell derivation
- ✓ payload-binder consistency
- ✓ secret / mode consistency

a single proof object

client report = proof + public binders + encrypted payload package

Stage 2: Shuffler-Side Verification

1

chain-state check

predecessor commitment,
timestamp monotonicity



2

public-signal consistency

enrollment state, anchor, mode



3

ciphertext-package integrity check

transport consistency



4

verify proof

trajectory and state consistency



5

check payload binders

payload consistent with
the committed primary



6

state update / hold / blacklist

replay / identity / drift

invalid updates are rejected
before secure aggregation



Payload swapping is rejected if the
payload binders are inconsistent
or the proof fails.

Verification & Security Semantics



continuity
soundness



zero
knowledge



payload-binding
integrity



public-output DP

The proof enforces
trajectory continuity
and payload-binding
consistency.

Public binders tie the
committed primary
to the payload
contents.

Transcript integrity
protects the
encrypted package
in transit.

Stage 3: Dual Secure Aggregation & DP Release

Aggregation

Aggregator A

- decrypt payload
- re-check format / context
- add valid share

A and R
do not collude

Aggregator R

- decrypt payload
- re-check format / context
- add valid share

Decoder
reconstruct
aggregate

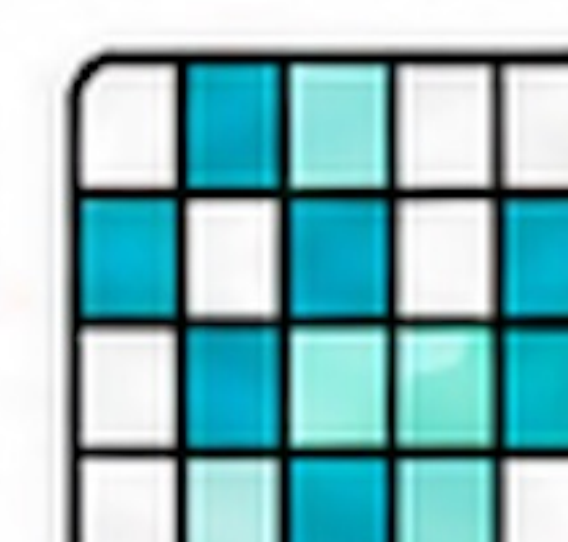
DP Release



SVT
threshold
filtering



Laplace
perturbation



Published
DP heatmap

- only accepted reports reach aggregation
- public-output DP applies only to the released heatmap
- A and R must not collude

accepted
reports
only